V2G Module Communication Protocol V1.3

Change of information registration form

Version	Reason for modification	Modification Notes	Modified by	Modification time
V1.0	New draft			2024/06/15
V1.1	New entry	1. Add reset command for DC side undervoltage in inverter mode 2. Add reset command for AC side undervoltage in off-grid mode 3. Add reset command for DC side undervoltage in rectification mode		2024/11/20
V1.2		AC phase mismatch is allowed by default		2024/12/11
V1.3		Modify the alarm status table		2025/2/21

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1 Overview

This protocol is applicable to NIUERA **AC/DC** Bidirectional module. XEV22KFGEU-FAC

Baud rate:125 Kbps

2 Working mode definition

NIUERA AC/DC The bidirectional module has three working modes: Grid-connected, Off-grid, and Rectifier mode.

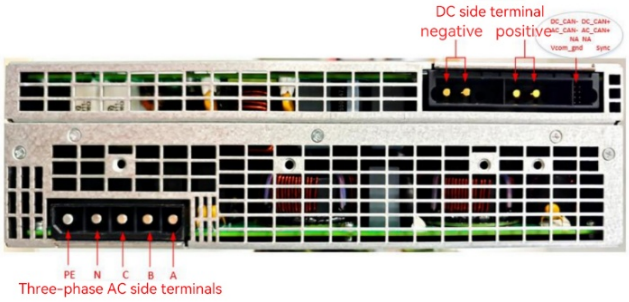
Rectifier mode: defined as charging from the AC side to the DC side, and the DC side current value is positive;

Grid-connected and Off-grid modes: defined as discharging from the DC side to the AC side, and the DC side current value is negative.;

[Note1] :

AC Side: Connect to one side of the three-phase AC outlet. See the figure below for the specific interface.

DC Side: Connect to one side of the DC side outlet. See the figure below for specific interfaces.



3 Frame Format Definition

3.1 Frame Format

Frame is the basic unit of information transmission. The CAN 2.0B frame format is shown in the following table :

illustrate	Code
Start of Frame	Sof(1bit)
Arbitration Domain	**Identifier(11bit)**
	SRR
	IDE
	Identifier(18bit)
	RTR
Control Code	Reseal(2 bits)
	Data Len(4 bits)
Data Domain	**Data(8 bytes)**
Checksum	CRC(2bits)
Terminator	(7 bits)

The controllable part used by actual users:

Identifier Field	Data Domain			
29-Bit	1byte	1byte		1byte
Frame Identifier Header	Data (1-8byte)			

3.2 Frame Identifier - 29 bits

28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PROTNO (9 bits)									PTP	DSTADDR (8 bits)								SRCADDR (8 bits)								Group		

3.2.1 PROTNO

Default: PROTNO= 0x061 Indicates bidirectional **AC / DC** Communication protocol number between module and monitoring

3.2.2 PTP

PTP =1, point-to-point communication;

PTP =0, broadcast communication;

3.2.3 DSTADDR

Destination Address:

Power module address range: 00~60;

The monitoring address is fixed at: 0xF0;

Broadcast address: 0xFF ;

Intra-group broadcast address: 0xFE

3.2.4 SRCADDR

Source address:

In all communication types, bits 3 to 10 are used to indicate the address of the message source on the bus.

Power module address range: 0~60

The monitoring address is fixed at: 0xF0

3.2.5 Group

Module group number: 0~7

3.3 Data Domain

3.3.1 Setting module parameters

Used to set the module voltage,current, current limit, power on, power off.

Transmit Frame	**Data Field**	**Format**	
Byte0	Byte1	Byte2~Byte3	Byte4~Byte7
Function code	Reserve	Register Number	Data
03	00	See **Table 1**	Data to be set (refer to **Table 1**)

(1) Example: Setting DC-side current in Rectifier mode.

03	00	00 79	00 00 1F 40

Note: The integer data corresponding to 0x00001F40 is: 8000, 8000 = 8*1000, which means setting the DC side current value to8A.

(2) Example: Setting the module to Power on

03	00	00 30	00 00 00 00

(3) Example: Setting module to Power off

03	00	00 30	00 01 00 00

3.3.2 Read module data

Transmit Frame	**Data Field**	**Format**	
Byte0	Byte1	Byte2~Byte3	Byte4~Byte7
Function code	Keep default	Register Number	Keep default
10	00	See **Table 1**	00 00 00 00

Response Field	**Data Field**	**Format**	
Byte0	Byte1	Byte2~Byte3	Byte4~Byte7
The type of returned data	Error Code	Register Number	Keep default
42: The returned value is an integer	F0: Normal Other: Fault, this frame is discarded.	See **Table 1**	Returned data

(1) Example: Reading DC voltage and current

Monitor sending:

10	00	00 0E	00 00 00 00

Module response:

42	F0	00 0E	0B B8 08 98

0x42: The returned data is in integer format

0xF0: Return frame is normal

0x000E: Register number

The integer format data corresponding to 0x0BB8 is: 3000, 3000=300*10, that is, the current DC side voltage is 300V.

The integer format data corresponding to 0x0BB8 is: 2200, 2200=2 2*100, that is, the current DC side current is 22A.

(2) Example: Read module status

Monitor sending:

10	00	00 40	00 00 00 00

Module response:

42	F0	00 40	00 00 01 00

0x42: The returned data is in integer format

0xF0: Return frame is normal

0x0040: Register number

0x00000100. See **table2**

4 Appendix Table

4.1 Data Type Parsing Table

Table 1

Register Address (VALUETYPE)	**Data Description** (RMP)	**Data Format**	**Format description**
0x0001	Read AC A Phase, phase voltage and phase current	Integer	BYTE4~BYTE5:A Phase voltage (unit:0.1V) BYTE6~BYTE7:A Phase current (unit: 0.01 A)
0x0002	Read AC B phase, phase voltage and phase current	Integer	BYTE4~BYTE5: Phase B voltage (unit: 0.1V) BYTE6~BYTE7: Phase B current (unit: 0.01A)
0x0003	Read AC C phase, phase voltage and phase current	Integer	BYTE4~BYTE5: Phase C voltage (unit: 0.1V) BYTE6~BYTE7: Phase C current (unit: 0.01A)
0x0004	Read AC AB line voltage	Integer	Unit: 0.1V

0x0005	Reading AC BC Line Voltage	Integer	Unit: 0.1V
0x0006	Read AC CA line voltage	Integer	Unit: 0.1V
0x0007	Read phase A, active power and reactive power	Integer	BYTE4~BYTE5: Phase A active power (unit: W) BYTE6~BYTE7: Phase A reactive power (unit: Var)
0x0008	Read phase B, active power and reactive power	Integer	BYTE4~BYTE5: Phase B active power (unit: W) BYTE6~BYTE7: Phase B reactive power (unit: Var)
0x0009	Read phase C, active power and reactive power	Integer	BYTE4~BYTE5: Phase C active power (unit: W) BYTE6~BYTE7: Phase C reactive power (unit: Var)
0x000A	Read AC frequency	Integer	Unit: mHz
0x000B	Read module panel (ambient) temperature	Integer	Unit: 0.001℃
0x000C	Read total active power	Integer	Unit: W
0x000D	Read total reactive power	Integer	Unit：Var
0x000E	Read total apparent power	Integer	Unit：VA
0x000F	Read DC side, voltage and current	Integer	BYTE4~BYTE5: DC side voltage (unit: 0.1V) BYTE6~BYTE7: DC side current, rectifier mode returns a positive number, inverter mode returns a negative number (unit: 0.01A)
0x0011	Read the rated output power	Integer	Unit：0.001W
0x0017	Set the module's operating altitude	Integer	Unit: m; range is 1000~5000. If the altitude is lower than 1000m, no setting is required (no derating). If the altitude is higher than 5000m, set 5000. The setting value is saved after power off.
0x001E	Set Group Number	Integer	The lower three bits of byte7 (range 0~7) The remaining byte4~byte6 and the upper five bits of byte7 are zero.
0x001F	Set the module address allocation method	Integer	0x00000000: Automatic allocation 0x00010000: Dial setting (default)
0x0024	Set the AC side undervoltage reset in off-grid mode	Integer	0x00000000: Disable; 0x00010000: Reset
0x0025	Set the DC side undervoltage reset in rectification mode	Integer	0x00000000: Disable; 0x00010000: Reset
0x002F	Set the module working mode	Integer	0x00000000: Grid-connected mode (default) 0x00010000: Off-grid mode 0x00020000: Rectification mode Note: Switching is only possible when the device is powered off
0x0030	Set the module to power on or off	Integer	0x00010000: shutdown; 0x00000000: startup
0x0031	Set the module DC side overvoltage reset	Integer	0x00000000: Disable; 0x00010000: Reset
0x0040	Read the current alarm/status of the module	Integer	See Table 2

0x0042	Set the DC side undervoltage reset in inverter mode	Integer	0x00000000: Disable; 0x00010000: Reset
0x0043	Read module group address	Integer	BYTE4~BYTE5: module group number BYTE6~BYTE7: module address
0x0044	Set module short circuit reset	Integer	0x00000000: Disable; 0x00010000: Reset
0x004A	Read the current altitude setting	Integer	Default 1000m (unit: m)
0x0056	Read module software version number	Integer	The lower 16 bits of the returned data (byte6~byte7), the version number is based on the decimal number of the data
0x0077	Setting the DC link voltage	Integer	Unit: mV
0x0079	Set the DC current	Integer	Unit: mA (positive number in rectification mode, negative number in inverter mode)

0x0080	Set total active power on the AC side	Integer	BYTE4~BYTE5: Power control in grid-connected mode (0x0000: control DC side power (default); 0x0001: control AC side power) BYTE6~BYTE7: total active power on AC side (unit: W)
0x0081	Set total reactive power on the AC side	Integer	Unit: Var (effective when setting the reactive power command, inductive is positive and capacitive is negative)
0x0082	Set AC side power factor PF	Integer	Unit: 0.001 (effective when the PF setting command is used, inductive is positive and capacitive is negative)
0x0083	Set the reactive power type on the AC side	Integer	0x00A00000: Do not set the reactive power output function (default) 0x00A10000: Use the PF setting command 0x00A20000: Use the reactive power setting command
0x0084	Set the rated phase voltage and AC frequency	Integer	BYTE4~BYTE5: Rated phase voltage (unit: 0.1V) BYTE6~BYTE7: AC frequency (unit: mHz)
0x0085	Set whether the phase is wrong	Integer	0x00000000: Phase error is not allowed 0x00010000: Phase error is allowed (default)
0x0086	Setting up the island detection function	Integer	0x00000000: Enable island detection (default) 0x00010000: Disable island detection
0x0087	Set the DC undervoltage protection voltage	Integer	Unit: 0.1V
0x0088	Set the DC overvoltage protection voltage	Integer	Unit: 0.1V
0x0089	Set AC undervoltage protection voltage and time	Integer	BYTE4~BYTE5: protection voltage (unit: 0.1V) BYTE6~BYTE7: protection time (unit: 0.01s)

0x008A	Set AC overvoltage protection voltage and time	Integer	BYTE4~BYTE5: protection voltage (unit: 0.1V) BYTE6~BYTE7: protection time (unit: 0.01s)
0x008B	Set the frequency and time of the first-level AC underfrequency protection	Integer	BYTE4~BYTE5: Protection frequency (unit: 0.01Hz) BYTE6~BYTE7: Protection time (unit: 0.01s)
0x008C	Set the first-level AC overfrequency protection frequency and time	Integer	BYTE4~BYTE5: Protection frequency (unit: 0.01Hz) BYTE6~BYTE7: Protection time (unit: 0.01s)
0x008D	Set the secondary AC underfrequency protection frequency and time	Integer	BYTE4~BYTE5: Protection frequency (unit: 0.01Hz) BYTE6~BYTE7: Protection time (unit: 0.01s)
0x008E	Set the secondary AC over-frequency protection frequency and time	Integer	BYTE4~BYTE5: Protection frequency (unit: 0.01Hz) BYTE6~BYTE7: Protection time (unit:0.01s)

4.2 Module Alarm Status Table

0 Invalid: 1 Valid

Table 2 - Module Alarm Status	
Bit	Description
0	Module failure (red indicator light on)
1	Module protection (yellow indicator light on)
2	AC input phase loss
3	Module internal SCI communication failure
4	The AC side wiring is wrong phase
5	Islanding alarm 0: Normal; 1: Islanding
6	Internal busbar over-voltage or under-voltage
7	AC side undervoltage
8	AC side overvoltage
9	DC side overvoltage
10	DC side undervoltage
11	Phase lock error
12-13	Module working mode 0: Grid-connected, 1: Off-grid, 2: Rectifier
14	U1 Overcurrent protection

15	Fan failure
16	CAN communication failure
17	Unbalanced current between modules
18	Duplicate address
19	N line connected incorrectly
20	Discharge failure
21	U1 power on/off status 0: power on, 1: power off
22	U2 Power on/off status 0: power on, 1: power off
23	Module power limit
24	Temperature limit power
25	AC power limit
26	AC side underfrequency
27	AC side overfrequency
28	DC side short circuit
29	Air duct overheating
30	Module over temperature
31	Ambient temperature over temperature

5 Set Group Number and Address Command：

Add commands for setting the **group number** and **address assignment method**. After a successful setting, the group number and address assignment method will be updated immediately and saved to memory (retained after power loss). The ID in the returned data will be updated to the latest group number immediately.

1、Group number setting instructions：

There are two ways to set the group number: via a DIP switch or through communication settings. These correspond to the automatic allocation mode and the DIP switch address setting mode, when the module's address allocation mode is set to automatic, the DIP switch is used to set the group number. In this case, the communication setting function for the group number will be invalid, and any command sent to do so will return a "setting failed" response. The communication setting function for the group number is only valid when the module's address mode is set to DIP switch setting mode.

2、The specific design routine is as follows：

(1) Example: Set the module group number to 5 (0x05)

Set command (data field)：

03	00	00 1E	00 00 00 05

Set the success reply command：

42	F0	00 1E	00 00 00 05

Set failure reply command：

42	F2	00 1E	00 00 00 05

(2) Example: Set the module address allocation mode to dial mode

Set command (data field)：

03	00	00 1F	00 01 00 00

Set the success reply command：

42	F0	00 1F	00 01 00 00

Set failure reply command：

42	F2	00 1F	00 01 00 00

6 Altitude Derating Instructions

When the module works in a high altitude area, the monitoring can set the current working altitude value. The effective setting range is 1000 meters to 5000 meters. The module will perform different degrees of derating according to the set altitude.

Considering that the air duct of the system may be blocked during long-term operation, it is recommended to set the actual altitude value when the module works at an altitude greater than 1000 meters.

Example: Set the module operating altitude to 3000 meters

Set command (data field)：

03	00	00 17	00 00 0B B8

Set the success reply command：

42	F0	00 17	00 00 0B B8

Set failure reply command：

42	F2	00 17	00 00 0B B8

7 Setting module working mode description

The monitoring module can be set to work in the following modes: grid-connected, off-grid, and rectifier mode. The module must be in shutdown state when setting. The control method is as follows：

(1) Set the module to grid-connected mode

Set command (data field)：

03	00	00 2F	00 00 00 00

Set the success reply command：

42	F0	00 2F	00 00 00 00

Set failure reply command：

42	F2	00 2F	00 00 00 00

(2) Set the module to off-grid mode

Set command (data field)：

03	00	00 2F	00 01 00 00

Set the success reply command：

42	F0	00 2F	00 01 00 00

Set failure reply command：

42	F2	00 2F	00 01 00 00

(3) Set the module to rectifier mode

Set command (data field)：

03	00	00 2F	00 02 00 00

Set the success reply command：

42	F0	00 2F	00 02 00 00

Set failure reply command：

42	F2	00 2F	00 02 00 00

8 Instructions for Switching Power Control Mode in Grid-connected Mode

When the module works in grid-connected mode, the monitoring can control the input power on the DC side and the active power on the AC side. The module needs to be in shutdown state when setting. The control method is as follows：

(1) In grid-connected mode, set the AC side active power to 20000W

Set command (data field)：

03	00	00 80	00 01 4E 20

0x0001 The grid-connected mode controls the active power on the AC side.

The corresponding integer data of 0x4E20 is: 20000, which means that the active power value on the AC side is set to 20000W.

Set the success reply command：

42	F0	00 80	00 01 4E 20

Set failure reply command：

42	F2	00 80	00 01 4E 20

(2) In grid-connected mode, set the DC side current to negative 40A

Setting command (data field):

03	00	00 80	00 00 4E 20

0x0000 The grid-connected mode controls the DC input power. 0x4E20 Invalid byte.

Set the success reply command：

42	F0	00 80	00 00 4E 20

Set failure reply command：

42	F2	00 80	00 00 4E 20

Set command (data field)：

03	00	00 79	FF FF 63 C0

The integer data corresponding to 0xFFFF63C0 is: -40000, -40000=-40*1000, which means setting the DC side current value to -40A.

Set the success reply command：

42	F0	00 79	FF FF 63 C0

Set failure reply command：

42	F2	00 79	FF FF 63 C0